

graphPlot

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# Chapter 1

## Namespace Index

### 1.1 Namespace List

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## Chapter 2

# Class Index

### 2.1 Class List

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## Chapter 3

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## Chapter 4

# Namespace Documentation

### 4.1 postprocess Namespace Reference

#### Namespaces

- [data](#)
- [export](#)
- [io](#)
- [layout](#)
- [plots](#)
- [style](#)

### 4.2 postprocess.data Namespace Reference

#### Namespaces

- [data1d](#)
- [dataset](#)
- [mesh](#)

### 4.3 postprocess.data.data1d Namespace Reference

#### Classes

- class [Data1D](#)

### 4.4 postprocess.data.dataset Namespace Reference

#### Classes

- class [DataSet](#)

## 4.5 postprocess.data.mesh Namespace Reference

### Classes

- class [Mesh](#)

## 4.6 postprocess.export Namespace Reference

### Namespaces

- [latex](#)
- [pdftex](#)

## 4.7 postprocess.export.latex Namespace Reference

### Classes

- class [LaTeXTextRegistry](#)

## 4.8 postprocess.export.pdftex Namespace Reference

### Classes

- class [PDFTeXExporter](#)

## 4.9 postprocess.io Namespace Reference

### Namespaces

- [tabular](#)
- [vtp](#)

## 4.10 postprocess.io.tabular Namespace Reference

### Functions

- def [read\\_1d](#) (filename, x\_column=0, y\_column=1, delimiter=None, skiprows=0, comments="#", label=None, x\_label=None, y\_label=None, x\_unit=None, y\_unit=None)

### 4.10.1 Function Documentation

#### 4.10.1.1 read\_1d()

```
def postprocess.io.tabular.read_1d (
    filename,
    x_column = 0,
    y_column = 1,
    delimiter = None,
    skiprows = 0,
    comments = "#",
    label = None,
    x_label = None,
    y_label = None,
    x_unit = None,
    y_unit = None )
```

Read two-column 1D scientific data.

Parameters

-----

filename:  
Input file.

x\_column:  
Column containing x data.

y\_column:  
Column containing y data.

delimiter:  
Column delimiter.

None:  
whitespace separated

", " :  
CSV

skiprows:  
Number of rows to skip.

comments:  
Comment character.

## 4.11 postprocess.io.vtp Namespace Reference

### Classes

- class [VTPData](#)

### Functions

- def [read\\_vtp](#) (Union[str, Path] filename)

#### 4.11.1 Function Documentation

#### 4.11.1.1 read\_vtp()

```
def postprocess.io.vtp.read_vtp (
    Union[str, Path] filename )
```

## 4.12 postprocess.layout Namespace Reference

### Namespaces

- [colorbar](#)
- [colors](#)
- [dimensions](#)
- [figure](#)

## 4.13 postprocess.layout.colorbar Namespace Reference

### Functions

- def [add\\_colorbar](#) (figure, axes, mappable, Optional[str] label=None, ticks=None, orientation="vertical", pad=0.02, fraction=0.046)

### 4.13.1 Function Documentation

#### 4.13.1.1 add\_colorbar()

```
def postprocess.layout.colorbar.add_colorbar (
    figure,
    axes,
    mappable,
    Optional[str] label = None,
    ticks = None,
    orientation = "vertical",
    pad = 0.02,
    fraction = 0.046 )
```

Add a colorbar to a figure.

Parameters

-----

figure  
Matplotlib figure.

axes  
Matplotlib axes.

mappable  
Scalar mappable returned by a plot.

`label`  
Colorbar label.

`ticks`  
Optional colorbar ticks.

`orientation`  
"vertical" or "horizontal".

`pad`  
Distance between axes and colorbar.

`fraction`  
Fraction of axes occupied by colorbar.

## 4.14 postprocess.layout.colors Namespace Reference

### Classes

- class [ColorScale](#)

## 4.15 postprocess.layout.dimensions Namespace Reference

### Functions

- def [get\\_width](#) (name)

### Variables

- float [SINGLE\\_COLUMN\\_WIDTH](#) = 3.5
- float [DOUBLE\\_COLUMN\\_WIDTH](#) = 7.0

### 4.15.1 Function Documentation

#### 4.15.1.1 [get\\_width\(\)](#)

```
def postprocess.layout.dimensions.get_width (  
    name )
```

Return a standard publication figure width.

### 4.15.2 Variable Documentation

#### 4.15.2.1 DOUBLE\_COLUMN\_WIDTH

```
float postprocess.layout.dimensions.DOUBLE_COLUMN_WIDTH = 7.0
```

#### 4.15.2.2 SINGLE\_COLUMN\_WIDTH

```
float postprocess.layout.dimensions.SINGLE_COLUMN_WIDTH = 3.5
```

### 4.16 postprocess.layout.figure Namespace Reference

#### Classes

- class [FigureConfig](#)
- class [PublicationFigure](#)

### 4.17 postprocess.plots Namespace Reference

#### Namespaces

- [annotation](#)
- [contour](#)
- [geometry](#)
- [line](#)
- [mesh](#)
- [streamline](#)
- [vector](#)

### 4.18 postprocess.plots.annotation Namespace Reference

#### Classes

- class [AnnotationPlot](#)

### 4.19 postprocess.plots.contour Namespace Reference

#### Classes

- class [ContourPlot](#)

## 4.20 postprocess.plots.geometry Namespace Reference

### Classes

- class [GeometryPlot](#)

## 4.21 postprocess.plots.line Namespace Reference

### Classes

- class [LinePlot](#)

## 4.22 postprocess.plots.mesh Namespace Reference

### Classes

- class [MeshPlot](#)

## 4.23 postprocess.plots.streamline Namespace Reference

### Classes

- class [StreamlinePlot](#)

## 4.24 postprocess.plots.vector Namespace Reference

### Classes

- class [VectorPlot](#)

## 4.25 postprocess.style Namespace Reference

### Namespaces

- [colors](#)
- [legend](#)
- [lines](#)
- [markers](#)
- [publication](#)

## 4.26 postprocess.style.colors Namespace Reference

### Functions

- def [get\\_colors](#) (scheme)
- def [validate\\_palette](#) (colors)

### Variables

- list [COLORBLIND\\_COLORS](#)
- list [DEFAULT\\_COLORS](#)
- list [GRAYSCALE\\_COLORS](#)
- list [BLACKWHITE\\_COLORS](#)

### 4.26.1 Detailed Description

Color schemes for publication figures.

### 4.26.2 Function Documentation

#### 4.26.2.1 [get\\_colors\(\)](#)

```
def postprocess.style.colors.get_colors (
    scheme )
```

Return the requested color scheme.

Parameters  
-----

scheme : str or sequence  
    Available named schemes:

- default
- colorblind
- grayscale
- blackwhite

A list or tuple may also be supplied to define  
a custom categorical palette.

Here is the call graph for this function:





### 4.26.2.2 validate\_palette()

```
def postprocess.style.colors.validate_palette (
    colors )
```

Validate a custom categorical color palette.

Parameters

-----

colors : sequence

Sequence of Matplotlib-compatible colors.

Returns

-----

list

Validated copy of the palette.

Raises

-----

ValueError

If the palette is empty.

TypeError

If colors is not a sequence.

Here is the caller graph for this function:



## 4.26.3 Variable Documentation

### 4.26.3.1 BLACKWHITE\_COLORS

```
list postprocess.style.colors.BLACKWHITE_COLORS
```

**Initial value:**

```
1 = [
2     "#000000",
3 ]
```

#### 4.26.3.2 COLORBLIND\_COLORS

```
list postprocess.style.colors.COLORBLIND_COLORS
```

##### Initial value:

```
1 = [
2     "#0072B2", # blue
3     "#E69F00", # orange
4     "#009E73", # green
5     "#D55E00", # vermillion
6     "#CC79A7", # purple
7     "#56B4E9", # sky blue
8     "#F0E442", # yellow
9     "#000000", # black
10 ]
```

#### 4.26.3.3 DEFAULT\_COLORS

```
list postprocess.style.colors.DEFAULT_COLORS
```

##### Initial value:

```
1 = [
2     "#1f77b4",
3     "#ff7f0e",
4     "#2ca02c",
5     "#d62728",
6     "#9467bd",
7     "#8c564b",
8     "#e377c2",
9     "#7f7f7f",
10 ]
```

#### 4.26.3.4 GRAYSCALE\_COLORS

```
list postprocess.style.colors.GRAYSCALE_COLORS
```

##### Initial value:

```
1 = [
2     "#000000",
3     "#404040",
4     "#707070",
5     "#999999",
6     "#BFBFBF",
7     "#D9D9D9",
8 ]
```

## 4.27 postprocess.style.legend Namespace Reference

### Classes

- class [LegendManager](#)

#### 4.27.1 Detailed Description

Legend management for publication figures.

## 4.28 postprocess.style.lines Namespace Reference

### Functions

- def [get\\_linestyles](#) (scheme)

### Variables

- list [PUBLICATION\\_LINESTYLES](#)
- list [BLACKWHITE\\_LINESTYLES](#)

#### 4.28.1 Detailed Description

Line style schemes for publication figures.

#### 4.28.2 Function Documentation

##### 4.28.2.1 [get\\_linestyles\(\)](#)

```
def postprocess.style.lines.get_linestyles (  
    scheme )
```

Return the requested line-style sequence.

#### 4.28.3 Variable Documentation

##### 4.28.3.1 [BLACKWHITE\\_LINESTYLES](#)

```
list postprocess.style.lines.BLACKWHITE_LINESTYLES
```

##### Initial value:

```
1 = [  
2     "-",  
3     "--",  
4     "-.",  
5     ":",  
6 ]
```

#### 4.28.3.2 PUBLICATION\_LINESTYLES

```
list postprocess.style.lines.PUBLICATION_LINESTYLES
```

##### Initial value:

```
1 = [  
2     "--",  
3     "---",  
4     "-.",  
5     ":",  
6 ]
```

## 4.29 postprocess.style.markers Namespace Reference

### Functions

- def [get\\_markers](#) (scheme)

### Variables

- list [PUBLICATION\\_MARKERS](#)
- list [BLACKWHITE\\_MARKERS](#)

#### 4.29.1 Detailed Description

Marker schemes for publication figures.

#### 4.29.2 Function Documentation

##### 4.29.2.1 get\_markers()

```
def postprocess.style.markers.get_markers (  
    scheme )
```

Return the requested marker sequence.

#### 4.29.3 Variable Documentation

#### 4.29.3.1 BLACKWHITE\_MARKERS

```
list postprocess.style.markers.BLACKWHITE_MARKERS
```

**Initial value:**

```
1 = [  
2     "O",  
3     "S",  
4     "^",  
5     "D",  
6     "V",  
7     "<",  
8     ">",  
9     "P",  
10    "X",  
11 ]
```

#### 4.29.3.2 PUBLICATION\_MARKERS

```
list postprocess.style.markers.PUBLICATION_MARKERS
```

**Initial value:**

```
1 = [  
2     "O",  
3     "S",  
4     "^",  
5     "D",  
6     "V",  
7     "<",  
8     ">",  
9     "P",  
10    "X",  
11 ]
```

## 4.30 postprocess.style.publication Namespace Reference

### Classes

- class [PublicationStyle](#)

#### 4.30.1 Detailed Description

Central publication style definition.

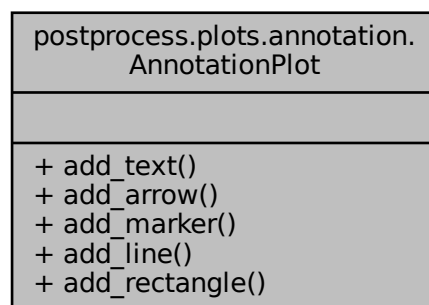


## Chapter 5

# Class Documentation

### 5.1 postprocess.plots.annotation.AnnotationPlot Class Reference

Collaboration diagram for postprocess.plots.annotation.AnnotationPlot:



#### Static Public Member Functions

- def `add_text` (axes, x, y, text, fontsize=10, ha="center", va="center", rotation=0, \*\*kwargs)
- def `add_arrow` (axes, start, end, color="black", linewidth=1.0, headwidth=8, headlength=10, \*\*kwargs)
- def `add_marker` (axes, position, marker="o", size=40, color="black", edgecolor=None, linewidth=1.0, \*\*kwargs)
- def `add_line` (axes, start, end, color="black", linewidth=1.0, linestyle="-", \*\*kwargs)
- def `add_rectangle` (axes, xy, width, height, edgecolor="black", facecolor="none", linewidth=1.0, linestyle="-", \*\*kwargs)

#### 5.1.1 Detailed Description

Scientific annotation utilities.

Provides simple publication-oriented primitives:

- text
- arrow
- marker
- line
- rectangle

## 5.1.2 Member Function Documentation

### 5.1.2.1 add\_arrow()

```
def postprocess.plots.annotation.AnnotationPlot.add_arrow (
    axes,
    start,
    end,
    color = "black",
    linewidth = 1.0,
    headwidth = 8,
    headlength = 10,
    ** kwargs ) [static]
```

Add an arrow between two data-coordinate positions.

Parameters

-----

start

(x, y) starting point.

end

(x, y) ending point.

### 5.1.2.2 add\_line()

```
def postprocess.plots.annotation.AnnotationPlot.add_line (
    axes,
    start,
    end,
    color = "black",
    linewidth = 1.0,
    linestyle = "-",
    ** kwargs ) [static]
```

Add a line between two data-coordinate positions.



### 5.1.2.3 add\_marker()

```
def postprocess.plots.annotation.AnnotationPlot.add_marker (
    axes,
    position,
    marker = "o",
    size = 40,
    color = "black",
    edgecolor = None,
    linewidth = 1.0,
    ** kwargs ) [static]
```

Add a marker at a data-coordinate position.

Parameters

-----

position  
    (x, y) position.

marker  
    Matplotlib marker style.

size  
    Marker size in points squared.

### 5.1.2.4 add\_rectangle()

```
def postprocess.plots.annotation.AnnotationPlot.add_rectangle (
    axes,
    xy,
    width,
    height,
    edgecolor = "black",
    facecolor = "none",
    linewidth = 1.0,
    linestyle = "--",
    ** kwargs ) [static]
```

Add a rectangle in data coordinates.

Parameters

-----

xy  
    Lower-left corner.

width  
    Rectangle width.

height  
    Rectangle height.

### 5.1.2.5 add\_text()

```
def postprocess.plots.annotation.AnnotationPlot.add_text (
    axes,
    x,
    y,
    text,
    fontsize = 10,
    ha = "center",
    va = "center",
    rotation = 0,
    ** kwargs ) [static]
```

Add text at a data-coordinate position.

Parameters

-----

axes

Matplotlib Axes.

x, y

Position in data coordinates.

text

Text or LaTeX expression.

fontsize

Font size in points.

ha

Horizontal alignment.

va

Vertical alignment.

rotation

Rotation angle in degrees.

The documentation for this class was generated from the following file:

- [src/postprocess/plots/annotation.py](#)

## 5.2 postprocess.layout.colors.ColorScale Class Reference

Collaboration diagram for postprocess.layout.colors.ColorScale:

| postprocess.layout.colors.<br>ColorScale                                         |
|----------------------------------------------------------------------------------|
| + levels<br>+ cmap<br>+ vmin<br>+ vmax<br>+ levels_array<br>+ colormap<br>+ norm |
| + __init__()<br>+ resolve()<br>+ limits()                                        |

### Public Member Functions

- def `__init__` (self, `levels`=30, `cmap`="viridis", `vmin`=None, `vmax`=None)
- def `resolve` (self, values)
- def `limits` (self)

### Public Attributes

- `levels`
- `cmap`
- `vmin`
- `vmax`
- `levels_array`
- `colormap`
- `norm`

### 5.2.1 Detailed Description

Defines the color scale for a scientific plot.

### 5.2.2 Constructor & Destructor Documentation

### 5.2.2.1 `__init__()`

```
def postprocess.layout.colors.ColorScale.__init__ (
    self,
    levels = 30,
    cmap = "viridis",
    vmin = None,
    vmax = None )
```

## 5.2.3 Member Function Documentation

### 5.2.3.1 `limits()`

```
def postprocess.layout.colors.ColorScale.limits (
    self )
```

### 5.2.3.2 `resolve()`

```
def postprocess.layout.colors.ColorScale.resolve (
    self,
    values )
```

Resolve the color scale using the supplied data.

## 5.2.4 Member Data Documentation

### 5.2.4.1 `cmap`

`postprocess.layout.colors.ColorScale.cmap`

### 5.2.4.2 `colormap`

`postprocess.layout.colors.ColorScale.colormap`

#### 5.2.4.3 levels

`postprocess.layout.colors.ColorScale.levels`

#### 5.2.4.4 levels\_array

`postprocess.layout.colors.ColorScale.levels_array`

#### 5.2.4.5 norm

`postprocess.layout.colors.ColorScale.norm`

#### 5.2.4.6 vmax

`postprocess.layout.colors.ColorScale.vmax`

#### 5.2.4.7 vmin

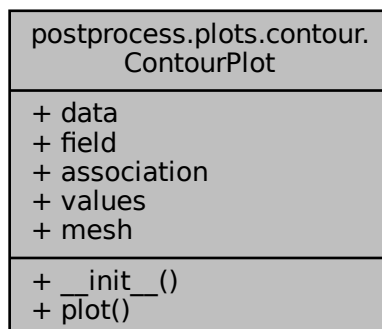
`postprocess.layout.colors.ColorScale.vmin`

The documentation for this class was generated from the following file:

- `src/postprocess/layout/colors.py`

## 5.3 postprocess.plots.contour.ContourPlot Class Reference

Collaboration diagram for `postprocess.plots.contour.ContourPlot`:



## Public Member Functions

- `def __init__ (self, data, field, association="cell")`
- `def plot (self, axes, scale=None, rasterize=True)`

## Public Attributes

- `data`
- `field`
- `association`
- `values`
- `mesh`

### 5.3.1 Detailed Description

Create a cell-centered scalar contour plot.

The field is associated with mesh cells.

### 5.3.2 Constructor & Destructor Documentation

#### 5.3.2.1 \_\_init\_\_()

```
def postprocess.plots.contour.ContourPlot.__init__ (
    self,
    data,
    field,
    association = "cell" )
```

Parameters

-----

`data`

Post-processing dataset.

`field`

Name of the scalar field.

`association`

Location of the field:  
"cell" or "point".

### 5.3.3 Member Function Documentation

### 5.3.3.1 plot()

```
def postprocess.plots.contour.ContourPlot.plot (
    self,
    axes,
    scale = None,
    rasterize = True )
```

Plot the scalar field.

Parameters  
-----

axes  
    Matplotlib axes.

levels  
    Number of discrete color levels.

cmap  
    Matplotlib colormap.

vmin, vmax  
    Color limits.

Returns  
-----

matplotlib.collections.PolyCollection  
    Contour collection.

## 5.3.4 Member Data Documentation

### 5.3.4.1 association

postprocess.plots.contour.ContourPlot.association

### 5.3.4.2 data

postprocess.plots.contour.ContourPlot.data

### 5.3.4.3 field

postprocess.plots.contour.ContourPlot.field

#### 5.3.4.4 mesh

`postprocess.plots.contour.ContourPlot.mesh`

#### 5.3.4.5 values

`postprocess.plots.contour.ContourPlot.values`

The documentation for this class was generated from the following file:

- `src/postprocess/plots/contour.py`

## 5.4 postprocess.data.data1d.Data1D Class Reference

Collaboration diagram for `postprocess.data.data1d.Data1D`:

| postprocess.data.data1d.<br>Data1D                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + x<br>+ y<br>+ label<br>+ x_label<br>+ y_label<br>+ x_unit<br>+ y_unit<br>+ metadata                                                                                       |
| + __init__()<br>+ size()<br>+ x_min()<br>+ x_max()<br>+ y_min()<br>+ y_max()<br>+ remove_invalid()<br>+ sort()<br>+ __len__()<br>+ __repr__()<br>+ scale_x()<br>+ scale_y() |



## Public Member Functions

- `def __init__ (self, x, y, label=None, x_label=None, y_label=None, x_unit=None, y_unit=None, metadata=None)`
- `def size (self)`
- `def x_min (self)`
- `def x_max (self)`
- `def y_min (self)`
- `def y_max (self)`
- `def remove_invalid (self)`
- `def sort (self)`
- `def __len__ (self)`
- `def __repr__ (self)`
- `def scale_x (self, scale, label=None, unit=None)`
- `def scale_y (self, scale, label=None, unit=None)`

## Public Attributes

- `x`
- `y`
- `label`
- `x_label`
- `y_label`
- `x_unit`
- `y_unit`
- `metadata`

### 5.4.1 Detailed Description

Generic one-dimensional scientific dataset.

Parameters  
-----

`x` : array-like  
Independent variable.

`y` : array-like  
Dependent variable.

`label` : str, optional  
Dataset label.

`x_label` : str, optional  
Label for x-axis.

`y_label` : str, optional  
Label for y-axis.

`x_unit` : str, optional  
Unit of x.

`y_unit` : str, optional  
Unit of y.

`metadata` : dict, optional  
Additional metadata.

## 5.4.2 Constructor & Destructor Documentation

### 5.4.2.1 `__init__()`

```
def postprocess.data.data1d.Data1D.__init__ (
    self,
    x,
    y,
    label = None,
    x_label = None,
    y_label = None,
    x_unit = None,
    y_unit = None,
    metadata = None )
```

## 5.4.3 Member Function Documentation

### 5.4.3.1 `__len__()`

```
def postprocess.data.data1d.Data1D.__len__ (
    self )
```

Here is the call graph for this function:



### 5.4.3.2 `__repr__()`

```
def postprocess.data.data1d.Data1D.__repr__ (
    self )
```

#### 5.4.3.3 remove\_invalid()

```
def postprocess.data.data1d.Data1D.remove_invalid (
    self )
```

Remove NaN and infinite values.

Returns

-----

Data1D

Cleaned dataset.

#### 5.4.3.4 scale\_x()

```
def postprocess.data.data1d.Data1D.scale_x (
    self,
    scale,
    label = None,
    unit = None )
```

Return a new Data1D with x scaled by `scale`.

$x_{\text{new}} = x / \text{scale}$

The original dataset is not modified.

Parameters

-----

scale : float  
Scaling reference.

label : str, optional  
New x-axis label.

unit : str, optional  
New x-axis unit.

Returns

-----

Data1D

Scaled dataset.

#### 5.4.3.5 scale\_y()

```
def postprocess.data.data1d.Data1D.scale_y (
    self,
    scale,
    label = None,
    unit = None )
```

Return a new Data1D with y scaled by 'scale'.

```
y_new = y / scale
```

The original dataset is not modified.

Parameters

-----

scale : float  
Scaling reference.

label : str, optional  
New y-axis label.

unit : str, optional  
New y-axis unit.

Returns

-----

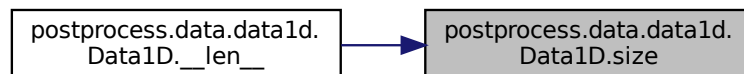
Data1D  
Scaled dataset.

#### 5.4.3.6 size()

```
def postprocess.data.data1d.Data1D.size (
    self )
```

Number of data points.

Here is the caller graph for this function:



#### 5.4.3.7 sort()

```
def postprocess.data.data1d.Data1D.sort (
    self )
```

Sort data according to x.

Returns

-----

Data1D  
Sorted dataset.

#### 5.4.3.8 x\_max()

```
def postprocess.data.data1d.Data1D.x_max (
    self )
```

#### 5.4.3.9 x\_min()

```
def postprocess.data.data1d.Data1D.x_min (
    self )
```

#### 5.4.3.10 y\_max()

```
def postprocess.data.data1d.Data1D.y_max (
    self )
```

#### 5.4.3.11 y\_min()

```
def postprocess.data.data1d.Data1D.y_min (
    self )
```

### 5.4.4 Member Data Documentation

#### 5.4.4.1 label

```
postprocess.data.data1d.Data1D.label
```

#### 5.4.4.2 metadata

```
postprocess.data.data1d.Data1D.metadata
```

#### 5.4.4.3 x

```
postprocess.data.data1d.Data1D.x
```

#### 5.4.4.4 x\_label

`postprocess.data.data1d.Data1D.x_label`

#### 5.4.4.5 x\_unit

`postprocess.data.data1d.Data1D.x_unit`

#### 5.4.4.6 y

`postprocess.data.data1d.Data1D.y`

#### 5.4.4.7 y\_label

`postprocess.data.data1d.Data1D.y_label`

#### 5.4.4.8 y\_unit

`postprocess.data.data1d.Data1D.y_unit`

The documentation for this class was generated from the following file:

- `src/postprocess/data/data1d.py`

## 5.5 postprocess.data.dataset.DataSet Class Reference

Collaboration diagram for `postprocess.data.dataset.DataSet`:

| postprocess.data.dataset.<br>DataSet                    |
|---------------------------------------------------------|
| + mesh<br>+ point_data<br>+ cell_data                   |
| + <code>__init__()</code><br>+ <code>get_field()</code> |

## Public Member Functions

- `def __init__ (self, mesh, point_data=None, cell_data=None)`
- `def get_field (self, name, association="cell")`

## Public Attributes

- `mesh`
- `point_data`
- `cell_data`

### 5.5.1 Detailed Description

Generic scientific dataset.

Contains geometry and associated fields.

### 5.5.2 Constructor & Destructor Documentation

#### 5.5.2.1 \_\_init\_\_()

```
def postprocess.data.dataset.DataSet.__init__ (
    self,
    mesh,
    point_data = None,
    cell_data = None )
```

### 5.5.3 Member Function Documentation

#### 5.5.3.1 get\_field()

```
def postprocess.data.dataset.DataSet.get_field (
    self,
    name,
    association = "cell" )
```

### 5.5.4 Member Data Documentation

#### 5.5.4.1 cell\_data

```
postprocess.data.dataset.DataSet.cell_data
```

#### 5.5.4.2 mesh

```
postprocess.data.dataset.DataSet.mesh
```

#### 5.5.4.3 point\_data

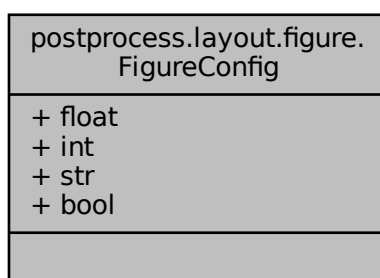
```
postprocess.data.dataset.DataSet.point_data
```

The documentation for this class was generated from the following file:

- [src/postprocess/data/dataset.py](#)

## 5.6 postprocess.layout.figure.FigureConfig Class Reference

Collaboration diagram for postprocess.layout.figure.FigureConfig:



### Static Public Attributes

- [float](#)
- [int](#)
- [str](#)
- [bool](#)



### 5.6.1 Detailed Description

Configuration for a publication figure.

### 5.6.2 Member Data Documentation

#### 5.6.2.1 bool

`postprocess.layout.figure.FigureConfig.bool` [static]

#### 5.6.2.2 float

`postprocess.layout.figure.FigureConfig.float` [static]

#### 5.6.2.3 int

`postprocess.layout.figure.FigureConfig.int` [static]

#### 5.6.2.4 str

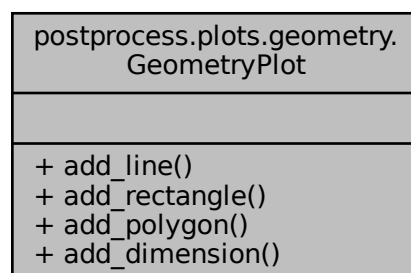
`postprocess.layout.figure.FigureConfig.str` [static]

The documentation for this class was generated from the following file:

- [src/postprocess/layout/figure.py](#)

## 5.7 postprocess.plots.geometry.GeometryPlot Class Reference

Collaboration diagram for `postprocess.plots.geometry.GeometryPlot`:



## Static Public Member Functions

- def [add\\_line](#) (axes, start, end, linewidth=1.0, linestyle="-", color="black", \*\*kwargs)
- def [add\\_rectangle](#) (axes, xy, width, height, linewidth=1.0, linestyle="-", edgecolor="black", facecolor="none", \*\*kwargs)
- def [add\\_polygon](#) (axes, points, linewidth=1.0, linestyle="-", edgecolor="black", facecolor="none", closed=True, \*\*kwargs)
- def [add\\_dimension](#) (axes, start, end, offset=0.15, text=None, linewidth=0.8, color="black", fontsize=9, text\_offset=0.05, arrowstyle="<->", \*\*kwargs)

### 5.7.1 Detailed Description

Utilities for plotting physical geometry and scientific reference geometry.

All coordinates are in data coordinates.

### 5.7.2 Member Function Documentation

#### 5.7.2.1 [add\\_dimension\(\)](#)

```
def postprocess.plots.geometry.GeometryPlot.add_dimension (
    axes,
    start,
    end,
    offset = 0.15,
    text = None,
    linewidth = 0.8,
    color = "black",
    fontsize = 9,
    text_offset = 0.05,
    arrowstyle = "<->",
    ** kwargs ) [static]
```

Add a dimension line between two points.

The dimension is offset perpendicular to the measured line.

Parameters

-----

start:

First measurement point.

end:

Second measurement point.

offset:

Perpendicular offset of the dimension line.

text:

Dimension label.

text\_offset:

Additional offset for the text.

arrowstyle:

Arrow style used for the dimension line.

### 5.7.2.2 add\_line()

```
def postprocess.plots.geometry.GeometryPlot.add_line (
    axes,
    start,
    end,
    linewidth = 1.0,
    linestyle = "-",
    color = "black",
    ** kwargs ) [static]
```

Draw a straight line between two points.

### 5.7.2.3 add\_polygon()

```
def postprocess.plots.geometry.GeometryPlot.add_polygon (
    axes,
    points,
    linewidth = 1.0,
    linestyle = "-",
    edgecolor = "black",
    facecolor = "none",
    closed = True,
    ** kwargs ) [static]
```

Draw a polygon.

Parameters

-----

points:

Sequence of (x, y) coordinates.

### 5.7.2.4 add\_rectangle()

```
def postprocess.plots.geometry.GeometryPlot.add_rectangle (
    axes,
    xy,
    width,
    height,
    linewidth = 1.0,
    linestyle = "-",
    edgecolor = "black",
    facecolor = "none",
    ** kwargs ) [static]
```

Draw a rectangle.

Parameters

-----

xy:

Lower-left corner.

width:

Width in data coordinates.

height:

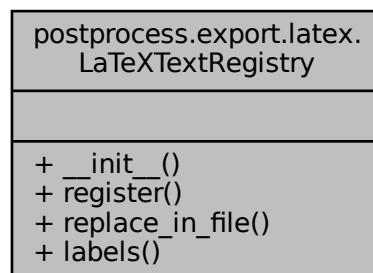
Height in data coordinates.

The documentation for this class was generated from the following file:

- [src/postprocess/plots/geometry.py](#)

## 5.8 postprocess.export.latex.LaTeXTextRegistry Class Reference

Collaboration diagram for postprocess.export.latex.LaTeXTextRegistry:



### Public Member Functions

- def `__init__`(self)
- def `register`(self, text)
- def `replace_in_file`(self, filename)
- def `labels`(self)

#### 5.8.1 Detailed Description

Stores LaTeX strings associated with temporary placeholders used during PDF+TeX export.

#### 5.8.2 Constructor & Destructor Documentation

### 5.8.2.1 `__init__()`

```
def postprocess.export.latex.LaTeXTextRegistry.__init__ (
    self )
```

## 5.8.3 Member Function Documentation

### 5.8.3.1 `labels()`

```
def postprocess.export.latex.LaTeXTextRegistry.labels (
    self )
```

Return the registered labels.

### 5.8.3.2 `register()`

```
def postprocess.export.latex.LaTeXTextRegistry.register (
    self,
    text )
```

Register a LaTeX string and return a unique plain-text placeholder.

### 5.8.3.3 `replace_in_file()`

```
def postprocess.export.latex.LaTeXTextRegistry.replace_in_file (
    self,
    filename )
```

Replace all registered placeholders in a PDF\_TeX file with their original LaTeX text.

The documentation for this class was generated from the following file:

- [src/postprocess/export/latex.py](#)

## 5.9 postprocess.style.legend.LegendManager Class Reference

Collaboration diagram for postprocess.style.legend.LegendManager:

|                                                                                           |
|-------------------------------------------------------------------------------------------|
| postprocess.style.legend.<br>LegendManager                                                |
| + reference_entries                                                                       |
| + <code>__init__</code> ()<br>+ <code>add_reference</code> ()<br>+ <code>render</code> () |

### Public Member Functions

- `def __init__` (self)
- `def add_reference` (self, label, handle=None)
- `def render` (self, axes, location="best", frameon=False, ncol=1, fontsize=None, bbox\_to\_anchor=None, columnspacing=1.5, handlelength=2.0, handletextpad=0.5, borderaxespad=0.5, \*\*kwargs)

### Public Attributes

- `reference_entries`

### 5.9.1 Detailed Description

Central manager for publication legends.

Handles data legends and explanatory/reference legend entries.

### 5.9.2 Constructor & Destructor Documentation

#### 5.9.2.1 `__init__`()

```
def postprocess.style.legend.LegendManager.__init__ (
    self )
```

## 5.9.3 Member Function Documentation

### 5.9.3.1 add\_reference()

```
def postprocess.style.legend.LegendManager.add_reference (
    self,
    label,
    handle = None )
```

Add an explanatory/reference legend entry.

Parameters

-----

label : str  
Text displayed in the legend.

handle : optional  
Matplotlib artist representing the entry.

### 5.9.3.2 render()

```
def postprocess.style.legend.LegendManager.render (
    self,
    axes,
    location = "best",
    frameon = False,
    ncol = 1,
    fontsize = None,
    bbox_to_anchor = None,
    columnspacing = 1.5,
    handlelength = 2.0,
    handletextpad = 0.5,
    borderaxespad = 0.5,
    ** kwargs )
```

Render the legend.

## 5.9.4 Member Data Documentation

### 5.9.4.1 reference\_entries

postprocess.style.legend.LegendManager.reference\_entries

The documentation for this class was generated from the following file:

- [src/postprocess/style/legend.py](#)

## 5.10 postprocess.plots.line.LinePlot Class Reference

Collaboration diagram for postprocess.plots.line.LinePlot:

| postprocess.plots.line.LinePlot                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + style<br>+ datasets<br>+ errorbars                                                                                                                                                                                                     |
| + <code>__init__()</code><br>+ <code>add()</code><br>+ <code>add_errorbar()</code><br>+ <code>plot()</code><br>+ <code>number_of_datasets()</code><br>+ <code>clear()</code><br>+ <code>legend()</code><br>+ <code>legend_table()</code> |

### Public Member Functions

- def `__init__` (self, [style](#)=None)
- def [add](#) (self, data, label=None, role=None, color=None, linewidth=1.5, linestyle=None, marker=None, markersize=4.0, markerfacecolor=None, markeredgecolor=None, markeredgewidth=1.0, markevery=None, alpha=1.0, zorder=3, \*\*kwargs)
- def [add\\_errorbar](#) (self, data, y\_error=None, x\_error=None, label=None, role=None, color=None, marker="o", markersize=4.0, markerfacecolor=None, markeredgecolor=None, markeredgewidth=1.0, linestyle="None", linewidth=1.0, capsize=3.0, capthick=None, elinewidth=None, alpha=1.0, zorder=4, errorevery=1, \*\*kwargs)
- def [plot](#) (self, axes)
- def [number\\_of\\_datasets](#) (self)
- def [clear](#) (self)
- def [legend](#) (self, axes, location="best", frameon=False, ncol=1, fontsize=None, groups=None, \*\*kwargs)
- def [legend\\_table](#) (self, axes, groups, location="lower center", bbox\_to\_anchor=(0.5, -0.05), fontsize=None, handle\_length=35, column\_spacing=35, row\_spacing=6)

### Public Attributes

- [style](#)
- [datasets](#)
- [errorbars](#)

#### 5.10.1 Detailed Description

Publication-quality 1D multi-dataset plot.

Supports ordinary lines, markers, and error bars.



## 5.10.2 Constructor & Destructor Documentation

### 5.10.2.1 \_\_init\_\_()

```
def postprocess.plots.line.LinePlot.__init__ (
    self,
    style = None )
```

## 5.10.3 Member Function Documentation

### 5.10.3.1 add()

```
def postprocess.plots.line.LinePlot.add (
    self,
    data,
    label = None,
    role = None,
    color = None,
    linewidth = 1.5,
    linestyle = None,
    marker = None,
    markersize = 4.0,
    markerfacecolor = None,
    markeredgecolor = None,
    markeredgewidth = 1.0,
    markevery = None,
    alpha = 1.0,
    zorder = 3,
    ** kwargs )
```

Add a Data1D dataset.

Parameters  
-----

role : str, optional  
Dataset role:

numerical  
analytical  
experimental  
reference

If specified, publication styling is  
automatically applied.

### 5.10.3.2 add\_errorbar()

```
def postprocess.plots.line.LinePlot.add_errorbar (
    self,
    data,
    y_error = None,
    x_error = None,
    label = None,
    role = None,
    color = None,
    marker = "o",
    markersize = 4.0,
    markerfacecolor = None,
    markeredgecolor = None,
    markeredgewidth = 1.0,
    linestyle = "None",
    linewidth = 1.0,
    capsize = 3.0,
    capthick = None,
    elinewidth = None,
    alpha = 1.0,
    zorder = 4,
    errorevery = 1,
    ** kwargs )
```

Add a dataset with x/y error bars.

Parameters

-----

data:  
Data1D object.

y\_error:  
Uncertainty in y.

Can be:  
scalar  
1D array  
(lower, upper)

x\_error:  
Uncertainty in x.

Can be:  
scalar  
1D array  
(lower, upper)

label:  
Legend label.

color:  
Error-bar and marker color.

marker:  
Marker style.

markersize:  
Marker size.

markerfacecolor:  
Marker fill color.

markeredgecolor:  
Marker edge color.

```

markedewidth:
    Marker edge width.

linestyle:
    Line style.

linewidth:
    Connecting line width.

capsize:
    Error-bar cap size.

capthick:
    Error-bar cap thickness.

elinewidth:
    Error-bar line width.

errorevery:
    Plot an error bar every N points.

kwargs:
    Additional Matplotlib errorbar options.

```

### 5.10.3.3 clear()

```

def postprocess.plots.line.LinePlot.clear (
    self )

```

### 5.10.3.4 legend()

```

def postprocess.plots.line.LinePlot.legend (
    self,
    axes,
    location = "best",
    frameon = False,
    ncol = 1,
    fontsize = None,
    groups = None,
    ** kwargs )

```

Create a legend for the plotted datasets.

Parameters

-----

axes  
Matplotlib Axes.

location  
Legend location.

frameon  
Whether to draw a legend frame.

ncol  
Number of legend columns.

```

fontsize
    Legend font size.

groups
    Optional grouped legend specification.

    Example
    -----
    groups=[
{
    "title": "Experiment",
    "labels": [
        "Experiment 1",
        "Experiment 2",
    ],
},
{
    "title": "Numerical",
    "labels": [
        "Case 1",
        "Case 2",
    ],
},
]

    If groups is None, a normal legend is
    created.

Returns
-----
matplotlib.legend.Legend or None

```

### 5.10.3.5 legend\_table()

```

def postprocess.plots.line.LinePlot.legend_table (
    self,
    axes,
    groups,
    location = "lower center",
    bbox_to_anchor = (0.5, -0.05),
    fontsize = None,
    handle_length = 35,
    column_spacing = 35,
    row_spacing = 6 )

```

Create a grouped table-style legend.

Layout  
-----

|        | Group 1 | Group 2 |  |
|--------|---------|---------|--|
| Case 1 | line    | line    |  |
| Case 2 | line    | line    |  |

Parameters  
-----

```

axes
    Matplotlib Axes.

groups
    Dictionary mapping group names to dictionaries
    of case names and Matplotlib artists.

```

```

    Example
    -----
    {
    "Temperature": {
        "Case 1": line_1,
        "Case 2": line_2,
    },
    "Pressure": {
        "Case 1": line_3,
        "Case 2": line_4,
    },
    }

    fontsize
        Text size.

    handle_length
        Length of the legend line representation.

    column_spacing
        Horizontal spacing between columns.

    row_spacing
        Vertical spacing between rows.

    Returns
    -----
    AnchoredOffsetbox
        The grouped legend artist.

```

### 5.10.3.6 number\_of\_datasets()

```

def postprocess.plots.line.LinePlot.number_of_datasets (
    self )

```

### 5.10.3.7 plot()

```

def postprocess.plots.line.LinePlot.plot (
    self,
    axes )

```

Plot all ordinary datasets and error-bar datasets.

Returns

-----

```

list
    Matplotlib artists.

```

## 5.10.4 Member Data Documentation

#### 5.10.4.1 datasets

```
postprocess.plots.line.LinePlot.datasets
```

#### 5.10.4.2 errorbars

```
postprocess.plots.line.LinePlot.errorbars
```

#### 5.10.4.3 style

```
postprocess.plots.line.LinePlot.style
```

The documentation for this class was generated from the following file:

- [src/postprocess/plots/line.py](#)

## 5.11 postprocess.data.mesh.Mesh Class Reference

Collaboration diagram for postprocess.data.mesh.Mesh:

| postprocess.data.mesh.Mesh                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------|
| + points<br>+ cells                                                                                                                        |
| + <code>__init__()</code><br>+ <code>n_points()</code><br>+ <code>n_cells()</code><br>+ <code>bounds()</code><br>+ <code>polygons()</code> |

### Public Member Functions

- `def __init__(self, points, cells)`
- `def n\_points(self)`
- `def n\_cells(self)`
- `def bounds(self)`
- `def polygons(self)`

## Public Attributes

- [points](#)
- [cells](#)

### 5.11.1 Detailed Description

Representation of a 2D polygonal mesh.

### 5.11.2 Constructor & Destructor Documentation

#### 5.11.2.1 `__init__()`

```
def postprocess.data.mesh.Mesh.__init__ (
    self,
    points,
    cells )
```

### 5.11.3 Member Function Documentation

#### 5.11.3.1 `bounds()`

```
def postprocess.data.mesh.Mesh.bounds (
    self )
```

#### 5.11.3.2 `n_cells()`

```
def postprocess.data.mesh.Mesh.n_cells (
    self )
```

#### 5.11.3.3 `n_points()`

```
def postprocess.data.mesh.Mesh.n_points (
    self )
```

#### 5.11.3.4 polygons()

```
def postprocess.data.mesh.Mesh.polygons (
    self )
```

### 5.11.4 Member Data Documentation

#### 5.11.4.1 cells

```
postprocess.data.mesh.Mesh.cells
```

#### 5.11.4.2 points

```
postprocess.data.mesh.Mesh.points
```

The documentation for this class was generated from the following file:

- [src/postprocess/data/mesh.py](#)

## 5.12 postprocess.plots.mesh.MeshPlot Class Reference

Collaboration diagram for postprocess.plots.mesh.MeshPlot:

| postprocess.plots.mesh.MeshPlot                    |
|----------------------------------------------------|
| + data<br>+ mesh                                   |
| + <code>__init__()</code><br>+ <code>plot()</code> |

### Public Member Functions

- def `__init__`(self, [data](#))
- def [plot](#)(self, axes, edgecolor="black", facecolor="none", linewidth=0.15, alpha=0.30)



## Public Attributes

- [data](#)
- [mesh](#)

### 5.12.1 Detailed Description

Plot the original computational mesh.

### 5.12.2 Constructor & Destructor Documentation

#### 5.12.2.1 `__init__()`

```
def postprocess.plots.mesh.MeshPlot.__init__ (
    self,
    data )
```

### 5.12.3 Member Function Documentation

#### 5.12.3.1 `plot()`

```
def postprocess.plots.mesh.MeshPlot.plot (
    self,
    axes,
    edgecolor = "black",
    facecolor = "none",
    linewidth = 0.15,
    alpha = 0.30 )
```

Plot the original mesh.

Parameters

-----

axes

Matplotlib axes.

edgecolor

Color of the cell boundaries.

facecolor

Face color of the cells.

Use "none" to keep the underlying  
contour visible.

linewidth

Width of the cell boundaries.

alpha

Transparency of the mesh.

Returns

-----

PolyCollection

Matplotlib mesh collection.

## 5.12.4 Member Data Documentation

### 5.12.4.1 data

`postprocess.plots.mesh.MeshPlot.data`

### 5.12.4.2 mesh

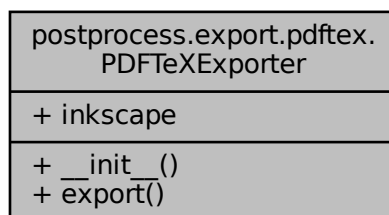
`postprocess.plots.mesh.MeshPlot.mesh`

The documentation for this class was generated from the following file:

- `src/postprocess/plots/mesh.py`

## 5.13 `postprocess.export.pdfTeX.PDFTeXExporter` Class Reference

Collaboration diagram for `postprocess.export.pdfTeX.PDFTeXExporter`:



### Public Member Functions

- `def __init__(self, inkscape="inkscape")`
- `def export(self, Union[str, Path] svg_file, Union[str, Path] output_file)`

### Public Attributes

- [inkscape](#)

### 5.13.1 Detailed Description

Export an SVG figure to PDF + PDF\_TeX using Inkscape.

### 5.13.2 Constructor & Destructor Documentation

#### 5.13.2.1 `__init__()`

```
def postprocess.export.pdf.tex.PDFTeXExporter.__init__ (
    self,
    inkscape = "inkscape" )
```

### 5.13.3 Member Function Documentation

#### 5.13.3.1 `export()`

```
def postprocess.export.pdf.tex.PDFTeXExporter.export (
    self,
    Union[str, Path] svg_file,
    Union[str, Path] output_file )
```

Export SVG to PDF + PDF\_TeX.

Parameters

-----

svg\_file  
Input SVG file.

output\_file  
PDF output filename.

Returns

-----

tuple  
PDF filename and PDF\_TeX filename.

### 5.13.4 Member Data Documentation

#### 5.13.4.1 `inkscape`

postprocess.export.pdf.tex.PDFTeXExporter.inkscape

The documentation for this class was generated from the following file:

- `src/postprocess/export/pdf.tex.py`

## 5.14 postprocess.layout.figure.PublicationFigure Class Reference

Collaboration diagram for postprocess.layout.figure.PublicationFigure:

| postprocess.layout.figure.<br>PublicationFigure                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + config<br>+ latex_registry<br>+ nrows<br>+ ncols<br>+ axes                                                                                                                                                          |
| + __init__()<br>+ panel()<br>+ set_common_xlabel()<br>+ set_common_ylabel()<br>+ label_panels()<br>+ adjust_layout()<br>+ add_shared_colorbar()<br>+ set_labels()<br>+ set_title()<br>+ set_limits()<br>and 6 more... |

### Public Member Functions

- def `__init__` (self, `config`, `nrows`=1, `ncols`=1, `sharex`=False, `sharey`=False)
- def `panel` (self, row, column)
- def `set_common_xlabel` (self, label, x=0.5, y=0.02)
- def `set_common_ylabel` (self, label, x=0.02, y=0.5)
- def `label_panels` (self, labels=None, x=0.02, y=0.98, fontsize=None)
- def `adjust_layout` (self, left=None, right=None, bottom=None, top=None, wspace=None, hspace=None)
- def `add_shared_colorbar` (self, mappable, `axes`, label=None, orientation="vertical", fraction=0.046, pad=0.04, shrink=1.0)
- def `set_labels` (self, xlabel=None, ylabel=None)
- def `set_title` (self, title, \*\*kwargs)
- def `set_limits` (self, xlim=None, ylim=None)
- def `set_ticks` (self, xticks=None, yticks=None)
- def `add_colorbar` (self, mappable, label=None, ticks=None, orientation="vertical", pad=0.02, fraction=0.046)
- def `save` (self, filename)
- def `save_svg` (self, filename, pdftex=False)
- def `export` (self, filename, formats)
- def `close` (self)

## Public Attributes

- [config](#)
- [latex\\_registry](#)
- [nrows](#)
- [ncols](#)
- [axes](#)

### 5.14.1 Detailed Description

Wrapper around a Matplotlib figure.

Controls figure layout, axes, labels, colorbars, and export.

### 5.14.2 Constructor & Destructor Documentation

#### 5.14.2.1 `__init__()`

```
def postprocess.layout.figure.PublicationFigure.__init__ (
    self,
    config,
    nrows = 1,
    ncols = 1,
    sharex = False,
    sharey = False )
```

### 5.14.3 Member Function Documentation

#### 5.14.3.1 `add_colorbar()`

```
def postprocess.layout.figure.PublicationFigure.add_colorbar (
    self,
    mappable,
    label = None,
    ticks = None,
    orientation = "vertical",
    pad = 0.02,
    fraction = 0.046 )
```

Add a colorbar.

Colorbar labels are registered for PDF+TeX export.

### 5.14.3.2 add\_shared\_colorbar()

```
def postprocess.layout.figure.PublicationFigure.add_shared_colorbar (
    self,
    mappable,
    axes,
    label = None,
    orientation = "vertical",
    fraction = 0.046,
    pad = 0.04,
    shrink = 1.0 )
```

Add one colorbar shared by multiple panels.

Parameters

-----

mappable : matplotlib.cm.ScalarMappable  
Contour/mappable returned by a plotting routine.

axes : list  
Axes sharing the colorbar.

label : str, optional  
Colorbar label.

orientation : {"vertical", "horizontal"}  
Colorbar orientation.

fraction : float  
Fraction of the axes width/height used by the colorbar.

pad : float  
Padding between the axes and colorbar.

shrink : float  
Colorbar length scaling.

Returns

-----

matplotlib.colorbar.Colorbar  
Created colorbar.

### 5.14.3.3 adjust\_layout()

```
def postprocess.layout.figure.PublicationFigure.adjust_layout (
    self,
    left = None,
    right = None,
    bottom = None,
    top = None,
    wspace = None,
    hspace = None )
```

Adjust spacing between figure panels.

#### 5.14.3.4 close()

```
def postprocess.layout.figure.PublicationFigure.close (
    self )
```

Close the Matplotlib figure.

#### 5.14.3.5 export()

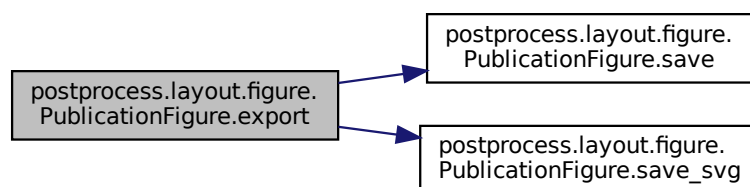
```
def postprocess.layout.figure.PublicationFigure.export (
    self,
    filename,
    formats )
```

Export the figure.

Supported formats:

```
png
pdf
svg
pdf_tex
```

Here is the call graph for this function:



#### 5.14.3.6 label\_panels()

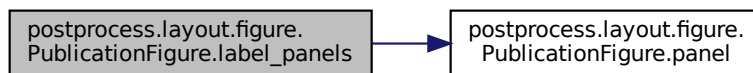
```
def postprocess.layout.figure.PublicationFigure.label_panels (
    self,
    labels = None,
    x = 0.02,
    y = 0.98,
    fontsize = None )
```

Add labels such as (a), (b), (c), ... to each panel.

#### Parameters

-----  
**labels** : list, optional  
 Custom panel labels.  
  
**x** : float, optional  
 Position inside each axes in axes coordinates.  
  
**y** : float, optional  
 Position inside each axes in axes coordinates.  
  
**fontsize** : float, optional  
 Panel-label font size.

Here is the call graph for this function:



#### 5.14.3.7 panel()

```
def postprocess.layout.figure.PublicationFigure.panel (
    self,
    row,
    column )
```

Return the axes corresponding to a panel.

#### Parameters

-----  
**row** : int  
 Zero-based row index.  
  
**column** : int  
 Zero-based column index.

#### Returns

-----  
`matplotlib.axes.Axes`  
 Requested panel.

Here is the caller graph for this function:



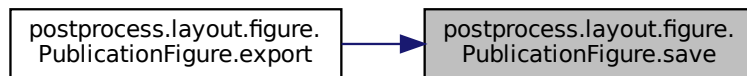


#### 5.14.3.8 save()

```
def postprocess.layout.figure.PublicationFigure.save (
    self,
    filename )
```

Save the figure normally.

Here is the caller graph for this function:



#### 5.14.3.9 save\_svg()

```
def postprocess.layout.figure.PublicationFigure.save_svg (
    self,
    filename,
    pdftex = False )
```

Save the figure as SVG.

When `pdftex=True`, LaTeX labels are temporarily replaced with plain-text placeholders so that Inkscape does not convert them into Unicode glyphs.

Here is the caller graph for this function:



#### 5.14.3.10 `set_common_xlabel()`

```
def postprocess.layout.figure.PublicationFigure.set_common_xlabel (
    self,
    label,
    x = 0.5,
    y = 0.02 )
```

Add a common x-axis label for the entire figure.

Parameters

-----

`label` : str  
Common x-axis label.

`x` : float, optional  
Horizontal figure coordinate.

`y` : float, optional  
Vertical figure coordinate.

#### 5.14.3.11 `set_common_ylabel()`

```
def postprocess.layout.figure.PublicationFigure.set_common_ylabel (
    self,
    label,
    x = 0.02,
    y = 0.5 )
```

Add a common y-axis label for the entire figure.

Parameters

-----

`label` : str  
Common y-axis label.

`x` : float, optional  
Horizontal figure coordinate.

`y` : float, optional  
Vertical figure coordinate.

#### 5.14.3.12 `set_labels()`

```
def postprocess.layout.figure.PublicationFigure.set_labels (
    self,
    xlabel = None,
    ylabel = None )
```

Set axis labels.

The original strings are retained so that they can be converted to LaTeX during PDF+TeX export.

#### 5.14.3.13 set\_limits()

```
def postprocess.layout.figure.PublicationFigure.set_limits (
    self,
    xlim = None,
    ylim = None )
```

Set axis limits.

#### 5.14.3.14 set\_ticks()

```
def postprocess.layout.figure.PublicationFigure.set_ticks (
    self,
    xticks = None,
    yticks = None )
```

Set axis tick locations.

#### 5.14.3.15 set\_title()

```
def postprocess.layout.figure.PublicationFigure.set_title (
    self,
    title,
    ** kwargs )
```

Set the axes title.

The title is registered for PDF+TeX export so that LaTeX expressions such as ```$\gamma$``` are preserved correctly.

### 5.14.4 Member Data Documentation

#### 5.14.4.1 axes

`postprocess.layout.figure.PublicationFigure.axes`

#### 5.14.4.2 config

`postprocess.layout.figure.PublicationFigure.config`

#### 5.14.4.3 latex\_registry

`postprocess.layout.figure.PublicationFigure.latex_registry`

#### 5.14.4.4 ncols

`postprocess.layout.figure.PublicationFigure.ncols`

#### 5.14.4.5 nrows

`postprocess.layout.figure.PublicationFigure.nrows`

The documentation for this class was generated from the following file:

- `src/postprocess/layout/figure.py`

## 5.15 postprocess.style.publication.PublicationStyle Class Reference

Collaboration diagram for `postprocess.style.publication.PublicationStyle`:

| postprocess.style.publication.<br>PublicationStyle                                         |
|--------------------------------------------------------------------------------------------|
| + color_scheme<br>+ line_scheme<br>+ marker_scheme<br>+ colors<br>+ linestyle<br>+ markers |
| + __init__()<br>+ color()<br>+ linestyle()<br>+ marker()<br>+ dataset_style()              |

## Public Member Functions

- def `__init__` (self, `color_scheme`="colorblind", `line_scheme`="publication", `marker_scheme`="publication")
- def `color` (self, index)
- def `linestyle` (self, index)
- def `marker` (self, index)
- def `dataset_style` (self, index, role=None)

## Public Attributes

- `color_scheme`
- `line_scheme`
- `marker_scheme`
- `colors`
- `linestyles`
- `markers`

### 5.15.1 Detailed Description

Central visual style for scientific figures.

Parameters

-----

`color_scheme` : str  
Color scheme.

`line_scheme` : str  
Line style scheme.

`marker_scheme` : str  
Marker scheme.

### 5.15.2 Constructor & Destructor Documentation

#### 5.15.2.1 `__init__`()

```
def postprocess.style.publication.PublicationStyle.__init__ (
    self,
    color_scheme = "colorblind",
    line_scheme = "publication",
    marker_scheme = "publication" )
```

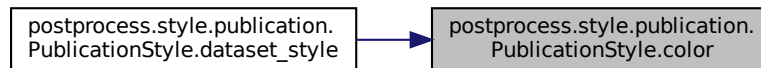
### 5.15.3 Member Function Documentation

### 5.15.3.1 color()

```
def postprocess.style.publication.PublicationStyle.color (
    self,
    index )
```

Return a color according to index.

Here is the caller graph for this function:



### 5.15.3.2 dataset\_style()

```
def postprocess.style.publication.PublicationStyle.dataset_style (
    self,
    index,
    role = None )
```

Return the visual style for a dataset.

Parameters

-----

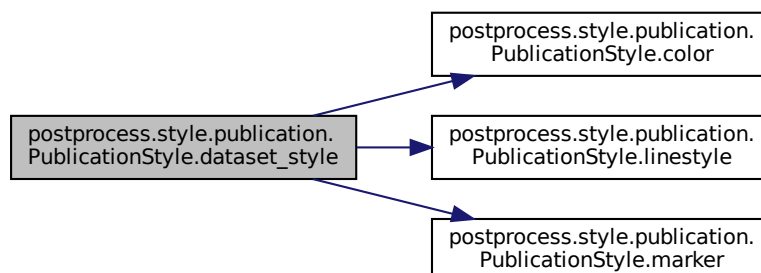
`index : int`  
Dataset index.

`role : str, optional`  
Dataset role.

Supported roles:

- numerical
- analytical
- experimental
- reference

Here is the call graph for this function:



### 5.15.3.3 linestyle()

```
def postprocess.style.publication.PublicationStyle.linestyle (
    self,
    index )
```

Return a line style according to index.

Here is the caller graph for this function:

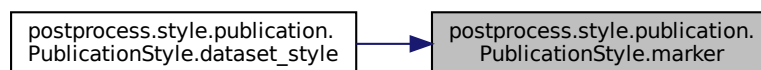


### 5.15.3.4 marker()

```
def postprocess.style.publication.PublicationStyle.marker (
    self,
    index )
```

Return a marker according to index.

Here is the caller graph for this function:



## 5.15.4 Member Data Documentation

#### 5.15.4.1 color\_scheme

`postprocess.style.publication.PublicationStyle.color_scheme`

#### 5.15.4.2 colors

`postprocess.style.publication.PublicationStyle.colors`

#### 5.15.4.3 line\_scheme

`postprocess.style.publication.PublicationStyle.line_scheme`

#### 5.15.4.4 linestyles

`postprocess.style.publication.PublicationStyle.linestyles`

#### 5.15.4.5 marker\_scheme

`postprocess.style.publication.PublicationStyle.marker_scheme`

#### 5.15.4.6 markers

`postprocess.style.publication.PublicationStyle.markers`

The documentation for this class was generated from the following file:

- [src/postprocess/style/publication.py](#)



## 5.16 postprocess.plots.streamline.StreamlinePlot Class Reference

Collaboration diagram for postprocess.plots.streamline.StreamlinePlot:

| postprocess.plots.streamline.<br>StreamlinePlot                                                            |
|------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>+ data</li> <li>+ field</li> <li>+ association</li> </ul>           |
| <ul style="list-style-type: none"> <li>+ <code>__init__()</code></li> <li>+ <code>plot()</code></li> </ul> |

### Public Member Functions

- `def __init__(self, data, field, association="auto")`
- `def plot(self, axes, n_seeds=25, seed_axis="y", seed_position=None, seed_margin=0.01, seed_↵  
points=None, integration_direction="forward", integrator_type=45, surface_streamlines=True, initial_step_↵  
length=0.1, min_step_length=0.01, max_step_length=0.5, max_steps=2000, max_length=None, terminal_↵  
_speed=1e-12, max_error=1e-6, interpolator_type="cell", color="black", linewidth=0.8, arrowsize=1.0,  
zorder=6)`

### Public Attributes

- `data`
- `field`
- `association`

#### 5.16.1 Detailed Description

Scientific streamline visualization using the VTK streamline integrator through PyVista.

Supports:

- point or cell vector fields
- explicit seed points
- regularly distributed seeds
- forward/backward/both integration
- RK2/RK4/RK45 integration
- surface-constrained integration
- physical maximum streamline length

#### 5.16.2 Constructor & Destructor Documentation

### 5.16.2.1 `__init__()`

```
def postprocess.plots.streamline.StreamlinePlot.__init__ (
    self,
    data,
    field,
    association = "auto" )
```

## 5.16.3 Member Function Documentation

### 5.16.3.1 `plot()`

```
def postprocess.plots.streamline.StreamlinePlot.plot (
    self,
    axes,
    n_seeds = 25,
    seed_axis = "y",
    seed_position = None,
    seed_margin = 0.01,
    seed_points = None,
    integration_direction = "forward",
    integrator_type = 45,
    surface_streamlines = True,
    initial_step_length = 0.1,
    min_step_length = 0.01,
    max_step_length = 0.5,
    max_steps = 2000,
    max_length = None,
    terminal_speed = 1e-12,
    max_error = 1e-6,
    interpolator_type = "cell",
    color = "black",
    linewidth = 0.8,
    arrowsize = 1.0,
    zorder = 6 )
```

Generate and plot streamlines using VTK.

Parameters

-----

**axes:**  
Matplotlib Axes.

**n\_seeds:**  
Number of automatically generated seed points.

**seed\_axis:**  
Axis along which automatic seeds are distributed.

**seed\_position:**  
Position of the seed line.

**seed\_margin:**  
Fraction of the seed line excluded at both ends.

```

seed_points:
    Explicit list/array of seed coordinates.
    If provided, this overrides automatic seeding.

integration_direction:
    "forward", "backward", or "both".

integrator_type:
    2  -> Runge-Kutta 2
    4  -> Runge-Kutta 4
    45 -> Runge-Kutta 45

surface_streamlines:
    Constrain integration to the 2-D surface.

initial_step_length:
    Initial integration step.

min_step_length:
    Minimum adaptive step.

max_step_length:
    Maximum adaptive step.

max_steps:
    Maximum number of integration steps.

max_length:
    Maximum physical streamline length.

terminal_speed:
    Integration terminates below this speed.

max_error:
    Maximum RK45 integration error.

interpolator_type:
    "point" or "cell".

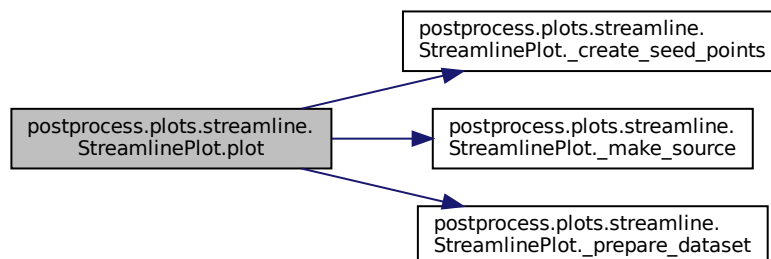
color:
    Streamline color.

linewidth:
    Streamline width.

arrowsize:
    Direction-arrow size.

```

Here is the call graph for this function:



## 5.16.4 Member Data Documentation

### 5.16.4.1 association

`postprocess.plots.streamline.StreamlinePlot.association`

### 5.16.4.2 data

`postprocess.plots.streamline.StreamlinePlot.data`

### 5.16.4.3 field

`postprocess.plots.streamline.StreamlinePlot.field`

The documentation for this class was generated from the following file:

- `src/postprocess/plots/streamline.py`

## 5.17 `postprocess.plots.vector.VectorPlot` Class Reference

Collaboration diagram for `postprocess.plots.vector.VectorPlot`:

| <code>postprocess.plots.vector.<br/>VectorPlot</code>                     |
|---------------------------------------------------------------------------|
| + <code>data</code><br>+ <code>field</code><br>+ <code>association</code> |
| + <code>__init__()</code><br>+ <code>plot()</code>                        |

### Public Member Functions

- `def __init__(self, data, field, association="auto")`
- `def plot(self, axes, density=20, scale=1.0, width=0.002, color="black", alpha=1.0, normalize=False, magnitude_color=False, cmap="viridis", pivot="mid", zorder=5, **kwargs)`

## Public Attributes

- [data](#)
- [field](#)
- [association](#)

### 5.17.1 Detailed Description

Plot 2D vector fields from a PyVista dataset.

Supports point-data and cell-data vector fields.

### 5.17.2 Constructor & Destructor Documentation

#### 5.17.2.1 `__init__()`

```
def postprocess.plots.vector.VectorPlot.__init__ (
    self,
    data,
    field,
    association = "auto" )
```

### 5.17.3 Member Function Documentation

#### 5.17.3.1 `plot()`

```
def postprocess.plots.vector.VectorPlot.plot (
    self,
    axes,
    density = 20,
    scale = 1.0,
    width = 0.002,
    color = "black",
    alpha = 1.0,
    normalize = False,
    magnitude_color = False,
    cmap = "viridis",
    pivot = "mid",
    zorder = 5,
    ** kwargs )
```

Plot a 2D vector field.

Parameters

-----

axes:

Matplotlib Axes.

density:

Approximate number of arrows along the smaller spatial dimension.

scale:

Quiver scale parameter.

width:

Arrow shaft width.

normalize:

Normalize all vectors to unit magnitude.

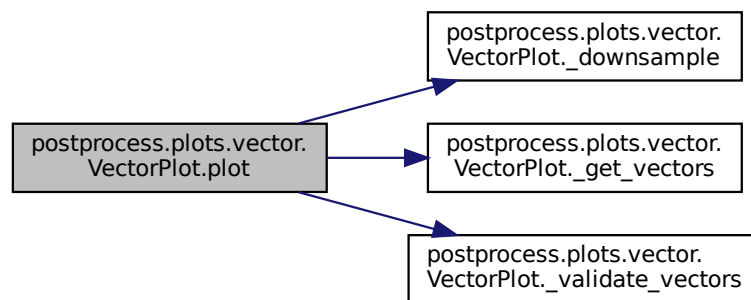
magnitude\_color:

Color arrows according to vector magnitude.

cmap:

Colormap used when magnitude\_color=True.

Here is the call graph for this function:



## 5.17.4 Member Data Documentation

### 5.17.4.1 association

`postprocess.plots.vector.VectorPlot.association`

### 5.17.4.2 data

`postprocess.plots.vector.VectorPlot.data`

### 5.17.4.3 field

`postprocess.plots.vector.VectorPlot.field`

The documentation for this class was generated from the following file:

- [src/postprocess/plots/vector.py](#)

## 5.18 postprocess.io.vtp.VTPData Class Reference

Collaboration diagram for `postprocess.io.vtp.VTPData`:

| postprocess.io.vtp.VTPData                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| + dataset<br>+ filename                                                                                                                                                                                                           |
| + <code>__init__()</code><br>+ <code>mesh()</code><br>+ <code>point_data()</code><br>+ <code>cell_data()</code><br>+ <code>bounds()</code><br>+ <code>n_points()</code><br>+ <code>n_cells()</code><br>+ <code>get_field()</code> |

### Public Member Functions

- `def __init__(self, dataset, filename)`
- `def mesh(self)`
- `def point_data(self)`
- `def cell_data(self)`
- `def bounds(self)`
- `def n_points(self)`
- `def n_cells(self)`
- `def get_field(self, name, association="cell")`

### Public Attributes

- `dataset`
- `filename`

### 5.18.1 Detailed Description

Data loaded from a VTP file.

### 5.18.2 Constructor & Destructor Documentation

#### 5.18.2.1 `__init__()`

```
def postprocess.io.vtp.VTPData.__init__ (
    self,
    dataset,
    filename )
```

### 5.18.3 Member Function Documentation

#### 5.18.3.1 `bounds()`

```
def postprocess.io.vtp.VTPData.bounds (
    self )
```

#### 5.18.3.2 `cell_data()`

```
def postprocess.io.vtp.VTPData.cell_data (
    self )
```

#### 5.18.3.3 `get_field()`

```
def postprocess.io.vtp.VTPData.get_field (
    self,
    name,
    association = "cell" )
```

#### 5.18.3.4 `mesh()`

```
def postprocess.io.vtp.VTPData.mesh (
    self )
```



### 5.18.3.5 n\_cells()

```
def postprocess.io.vtp.VTPData.n_cells (
    self )
```

### 5.18.3.6 n\_points()

```
def postprocess.io.vtp.VTPData.n_points (
    self )
```

### 5.18.3.7 point\_data()

```
def postprocess.io.vtp.VTPData.point_data (
    self )
```

## 5.18.4 Member Data Documentation

### 5.18.4.1 dataset

```
postprocess.io.vtp.VTPData.dataset
```

### 5.18.4.2 filename

```
postprocess.io.vtp.VTPData.filename
```

The documentation for this class was generated from the following file:

- [src/postprocess/io/vtp.py](#)



## Chapter 6

# File Documentation

### 6.1 src/postprocess/\_\_init\_\_.py File Reference

#### Namespaces

- [postprocess](#)

### 6.2 src/postprocess/data/\_\_init\_\_.py File Reference

#### Namespaces

- [postprocess.data](#)

### 6.3 src/postprocess/export/\_\_init\_\_.py File Reference

#### Namespaces

- [postprocess.export](#)

### 6.4 src/postprocess/io/\_\_init\_\_.py File Reference

#### Namespaces

- [postprocess.io](#)

### 6.5 src/postprocess/layout/\_\_init\_\_.py File Reference

#### Namespaces

- [postprocess.layout](#)

## 6.6 src/postprocess/plots/\_\_init\_\_.py File Reference

### Namespaces

- [postprocess.plots](#)

## 6.7 src/postprocess/style/\_\_init\_\_.py File Reference

### Namespaces

- [postprocess.style](#)

## 6.8 src/postprocess/data/data1d.py File Reference

### Classes

- class [postprocess.data.data1d.Data1D](#)

### Namespaces

- [postprocess.data.data1d](#)

## 6.9 src/postprocess/data/dataset.py File Reference

### Classes

- class [postprocess.data.dataset.DataSet](#)

### Namespaces

- [postprocess.data.dataset](#)

## 6.10 src/postprocess/data/mesh.py File Reference

### Classes

- class [postprocess.data.mesh.Mesh](#)

### Namespaces

- [postprocess.data.mesh](#)

## 6.11 src/postprocess/plots/mesh.py File Reference

### Classes

- class [postprocess.plots.mesh.MeshPlot](#)

### Namespaces

- [postprocess.plots.mesh](#)

## 6.12 src/postprocess/export/latex.py File Reference

### Classes

- class [postprocess.export.latex.LaTeXTextRegistry](#)

### Namespaces

- [postprocess.export.latex](#)

## 6.13 src/postprocess/export/pdftex.py File Reference

### Classes

- class [postprocess.export.pdftex.PDFTeXExporter](#)

### Namespaces

- [postprocess.export.pdftex](#)

## 6.14 src/postprocess/io/tabular.py File Reference

### Namespaces

- [postprocess.io.tabular](#)

### Functions

- def [postprocess.io.tabular.read\\_1d](#) (filename, x\_column=0, y\_column=1, delimiter=None, skiprows=0, comments="#", label=None, x\_label=None, y\_label=None, x\_unit=None, y\_unit=None)

## 6.15 src/postprocess/io/vtp.py File Reference

### Classes

- class [postprocess.io.vtp.VTPData](#)

### Namespaces

- [postprocess.io.vtp](#)

### Functions

- def [postprocess.io.vtp.read\\_vtp](#) (Union[str, Path] filename)

## 6.16 src/postprocess/layout/colorbar.py File Reference

### Namespaces

- [postprocess.layout.colorbar](#)

### Functions

- def [postprocess.layout.colorbar.add\\_colorbar](#) (figure, axes, mappable, Optional[str] label=None, ticks=None, orientation="vertical", pad=0.02, fraction=0.046)

## 6.17 src/postprocess/layout/colors.py File Reference

### Classes

- class [postprocess.layout.colors.ColorScale](#)

### Namespaces

- [postprocess.layout.colors](#)

## 6.18 src/postprocess/style/colors.py File Reference

### Namespaces

- [postprocess.style.colors](#)

## Functions

- def [postprocess.style.colors.get\\_colors](#) (scheme)
- def [postprocess.style.colors.validate\\_palette](#) (colors)

## Variables

- list [postprocess.style.colors.COLORBLIND\\_COLORS](#)
- list [postprocess.style.colors.DEFAULT\\_COLORS](#)
- list [postprocess.style.colors.GRAYSCALE\\_COLORS](#)
- list [postprocess.style.colors.BLACKWHITE\\_COLORS](#)

## 6.19 src/postprocess/layout/dimensions.py File Reference

### Namespaces

- [postprocess.layout.dimensions](#)

### Functions

- def [postprocess.layout.dimensions.get\\_width](#) (name)

### Variables

- float [postprocess.layout.dimensions.SINGLE\\_COLUMN\\_WIDTH](#) = 3.5
- float [postprocess.layout.dimensions.DOUBLE\\_COLUMN\\_WIDTH](#) = 7.0

## 6.20 src/postprocess/layout/figure.py File Reference

### Classes

- class [postprocess.layout.figure.FigureConfig](#)
- class [postprocess.layout.figure.PublicationFigure](#)

### Namespaces

- [postprocess.layout.figure](#)

## 6.21 src/postprocess/plots/annotation.py File Reference

### Classes

- class [postprocess.plots.annotation.AnnotationPlot](#)

## Namespaces

- [postprocess.plots.annotation](#)

## 6.22 src/postprocess/plots/contour.py File Reference

### Classes

- class [postprocess.plots.contour.ContourPlot](#)

### Namespaces

- [postprocess.plots.contour](#)

## 6.23 src/postprocess/plots/geometry.py File Reference

### Classes

- class [postprocess.plots.geometry.GeometryPlot](#)

### Namespaces

- [postprocess.plots.geometry](#)

## 6.24 src/postprocess/plots/line.py File Reference

### Classes

- class [postprocess.plots.line.LinePlot](#)

### Namespaces

- [postprocess.plots.line](#)

## 6.25 src/postprocess/plots/streamline.py File Reference

### Classes

- class [postprocess.plots.streamline.StreamlinePlot](#)



## Namespaces

- [postprocess.plots.streamline](#)

## 6.26 src/postprocess/plots/vector.py File Reference

### Classes

- class [postprocess.plots.vector.VectorPlot](#)

### Namespaces

- [postprocess.plots.vector](#)

## 6.27 src/postprocess/style/legend.py File Reference

### Classes

- class [postprocess.style.legend.LegendManager](#)

### Namespaces

- [postprocess.style.legend](#)

## 6.28 src/postprocess/style/lines.py File Reference

### Namespaces

- [postprocess.style.lines](#)

### Functions

- def [postprocess.style.lines.get\\_linestyles](#) (scheme)

### Variables

- list [postprocess.style.lines.PUBLICATION\\_LINESTYLES](#)
- list [postprocess.style.lines.BLACKWHITE\\_LINESTYLES](#)

## 6.29 src/postprocess/style/markers.py File Reference

### Namespaces

- [postprocess.style.markers](#)

## Functions

- def [postprocess.style.markers.get\\_markers](#) (scheme)

## Variables

- list [postprocess.style.markers.PUBLICATION\\_MARKERS](#)
- list [postprocess.style.markers.BLACKWHITE\\_MARKERS](#)

## 6.30 src/postprocess/style/publication.py File Reference

### Classes

- class [postprocess.style.publication.PublicationStyle](#)

### Namespaces

- [postprocess.style.publication](#)

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